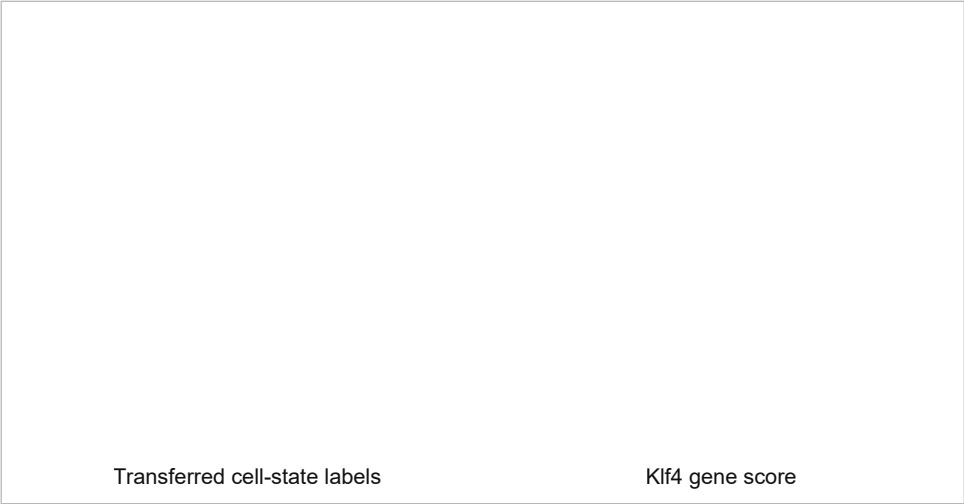
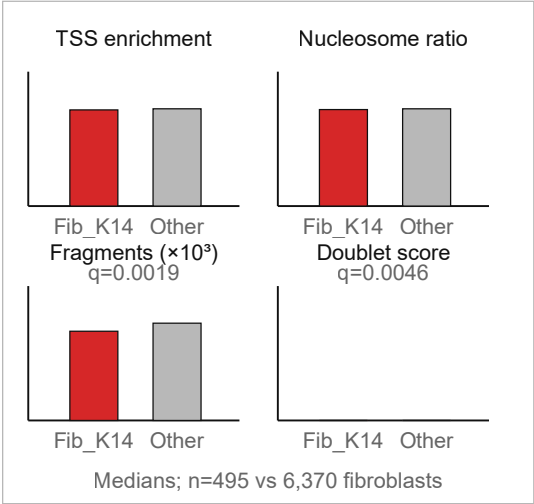


A RNA-anchored scATAC projection

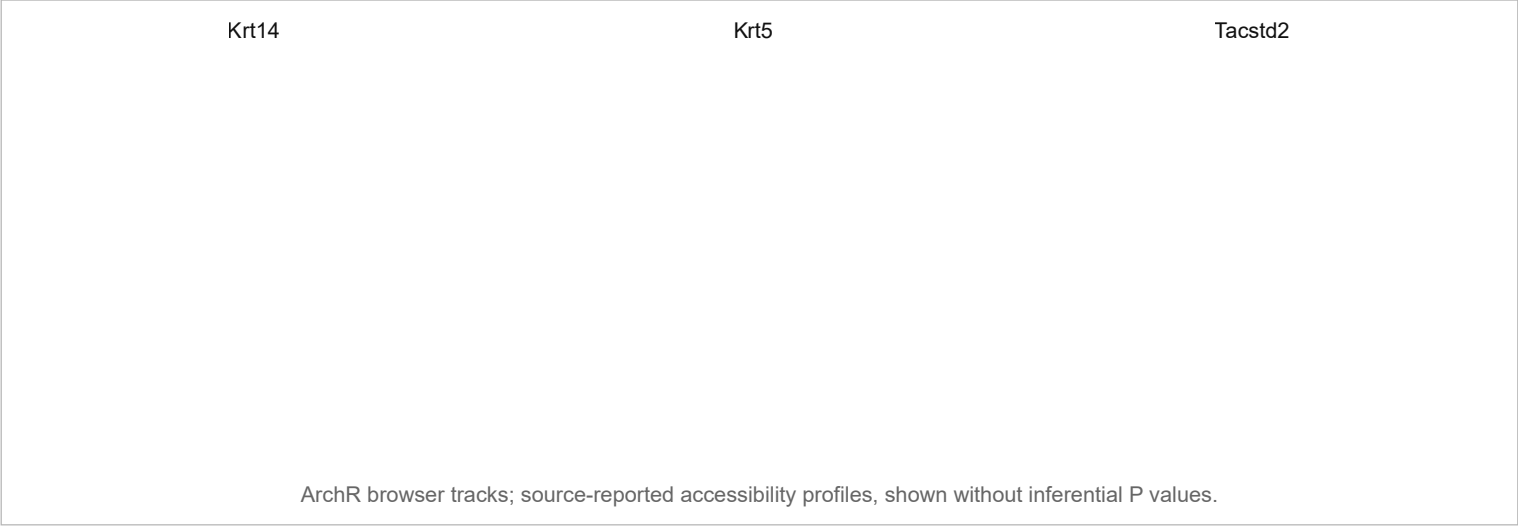


Label transfer projection; a joint RNA–ATAC coordinate object was not present in the supplied assets.

B scATAC-seq quality control



C Epithelial-associated chromatin accessibility



D KLF4 motif and footprinting audit

0 KLF4 motif hits

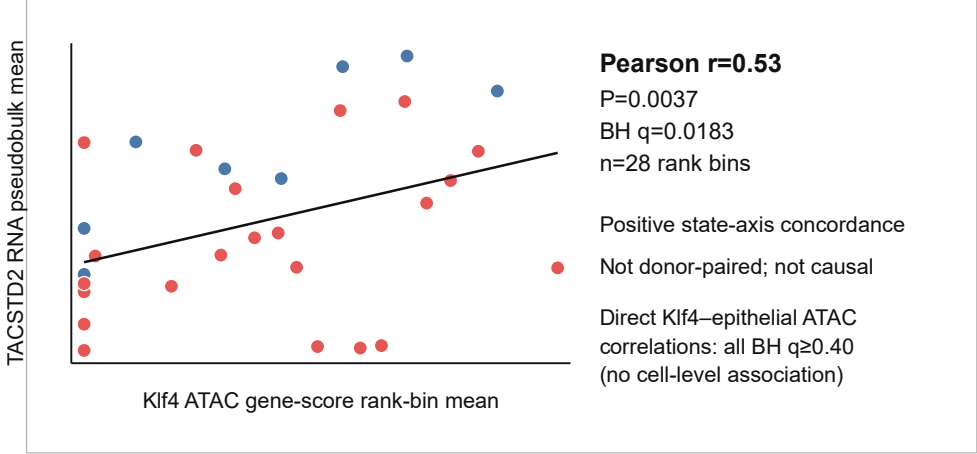
No valid KLF4 footprint can be computed from the supplied peak annotation.

Not a positive result

Do not claim KLF4 motif enrichment or footprinting until matched motif/peak data exist.

Input: KLF4_motif_hits_in_peakAnnotation.csv

E Multi-omics integration and evidence boundary



Supported

Fib_K14 marker accessibility is strongly elevated (Cliff's $\delta=0.79$; BH $q=5.6 \times 10^{-190}$).

Klf4-ATAC/TACSTD2-RNA rank-axis concordance is positive ($r=0.53$; BH $q=0.018$).

Not supported

Klf4 gene score is slightly lower in predicted Fib_K14 cells ($\delta=-0.057$; BH $q=0.049$).

No KLF4 motif hit or valid footprinting result.

Direct Klf4–epithelial ATAC correlations are null.

Interpretation boundary

Evidence supports a coordinated epithelial-like accessibility/expression state, but does not establish a direct KLF4 chromatin mechanism or fate conversion. Paired multiome or donor-matched validation is required.